The Effect of Forest Age on Machine Learning Evapotranspiration Generalisation

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ABSTRACT (200 words)
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Key words: Machine Learning, FLUXNET, Evapotranspiration.
Word count: 3000

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indica ted by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate refer encing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
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Literature Review

Introduction

Evapotranspiration

Terrestrial evapotranspiration (ET) is a key component of the global hydrological cycle describing the transition of water from liquid to gas at Earth's surface and further transfer to the atmosphere, and is a crucial component of the energy and water exchange between terrestrial ecosystems and the climate system. The term emphasises the combination of water vapourisation flux from (1) abiotic evaporation from soil and plant canopy surfaces, and (2) biotic transpiration through stomata to the atmosphere (Katul et al., 2012). ET can be generally described through:

$$ET = T + E_{Soil} + E_{Canopy} \tag{1}$$

where T refers to transpiration through plant stomata, E_{soil} to evaporation from the soil surface, and E_{Canopy} to evaporation from plant canopies (via rainfall interception) (Lawrence et al., 2007). The contribution of these variables to overall ET is not equal, with Lian et al. (2018) estimating a global T/ET ratio of 0.62±0.06, indicating plant transpiration as the leading component of ET worldwide. The partitioning of ET into E ($E_{soil} + E_{Canopy}$) and T within ecosystems is associated with "water use efficiency" and has use in irrigation management, where T is linked with plant productivity, while E does not directly contribute to production (Kool et al., 2014). Accurate partitioning estimates are also important in aiding carbon cycle projections given the coupled effect of transpiration and land-atmosphere carbon dioxide exchanges through stomata.

Forest Complexity

The magnitude of ET is governed by many environmental factors such as solar radiation, leaf area index (LAI), air temperature, vapour pressure deficit (VPD), soil moisture, and wind speed (Yang et al., 2023). Of these, Lian et al. (2018) identified LAI as the primary driver of T/ET spatial patterns, with Kool et al. (2014) suggesting that in systems with full canopy cover (and therefore high LAI) ET is often assumed to be "similar enough to T to permit correlation of ET with biomass". However, this assumption may break down in densely vegetated systems with multi-layer canopies such as forests, despite high LAI. In these systems, the 2-dimensional remote measure of LAI will not capture the vertical canopy complexity which drives rainfall interception and subsequent E_{Canopy}, implying that canopy cover indices may struggle to explain T/ET ratios in more complex vertical environments. This is reflected by Zhou et al. (2016) who report higher coefficients of determination between enhanced vegetation index (EVI) and T/ET in corn and soybean sites (0.84 and 0.82) with simple canopies, compared to evergreen needle leaf forests (0.37), a result possibly due to increased E_{Canopy} driving this ratio down. Hardiman et al. (2013) also demonstrate that while LAI may be stable in stands older than 50 years, canopy complexity continues to increase with age through >160 years of stand development and had a greater impact on annual net primary productivity (NPP) and overall resource use efficiency than LAI across all stands. Furthermore, LAI has shown to only explain around 20% of mean annual T/ET (Li et al., 2019), suggesting simple vegetation indices which don't take into account 3-dimensional complexity may be insufficient for predicting T/ET in structurally complex systems.

Oishi et al. (2020), in a chronosequence study on broadleaf forests in the US, report that interception of incoming rainfall strongly increases with age from ~2% (12 year old stand) to 20% (200 year old stand) despite only small differences in LAI, an effect in part due to branch architecture which varies with age. This study also reported that annual canopy transpiration (T) was over twice as high in the 35 year old stand compared to the 12 and 200 year old stands, with increasing canopy interception in the oldest stand largely offsetting the reduction in T such that total ET was similar between the two older stands, but remained lower in the youngest stand (Fig. 2). This demonstrates how age-driven effects in canopy architecture introduce complexity and can affect overall ET, as well as the relative contributions of T and E_{Canopy} which may vary significantly over time within stands.

Alongside canopy architecture, stomatal conductance (g_s) declines as trees grow taller due to gravitational reductions in leaf water potential and decreasing hydraulic conductance with increasing path length from soil to leaf (Ryan and Yoder, 1997). For example, Schäfer et al. (2000) demonstrated that the mean stomatal conductance of tree crowns decreased by approximately 60% over a 30m increase in tree height, and Hubbard et al. (1999) established a 44% drop in hydraulic conductance in old trees compared to young trees in a mixed age stand. As g_s directly influences T (Monteith, 1965), these age related changes introduce further complexity into ET dynamics and suggest that age could be a key source of variability in ET in forest ecosystems.

Forest Age

Forest age distributions are increasingly being affected by climate change-driven disturbances - such as fire, drought, insect attack, and pathogens - which are expected to continue in the coming decades (Solomon et al., 2009; Seidl et al., 2017). Some of these disturbances are able to completely change forest demography by causing widespread mortality (Dale et al., 2001), effectively resetting the age of the landscape. Land use changes also directly affect forest age distributions, for example the recovery of deforested areas in southern China following the implementation of forestation policies around the year 2000 has resulted in a complex pattern of forest age structure in recent years, and an increase in average tree cover from 21% in 1987 to 38% in 2016, leading to large areas now being dominated by young forests (Tong et al., 2020; Leng et al., 2024), with similar patterns of young forests due to recent disturbance found worldwide (Besnard et al., 2021).

As forests are exhibiting greater age heterogeneity, and with climate change increasing the severity and frequency of potential forest disturbances worldwide, stand age may represent an increasingly important variable for ET estimation in forests, capturing structural complexity and linking this to ET beyond what simple vegetation indices such as LAI are capable of. Yet whether these age-driven changes in forests such as canopy complexity and alterations to g_s are a large enough source of ET variability to affect the predictions made by current ET models which do not take age into account remains largely unexplored.

Evapotranspiration Prediction ~1400 Words

Contemporary ET Prediction Methods

FLUXNET

Machine Learning Models for ET Prediction

Machine Learning Applicability ~800 Words
(What is the case for using ML here?)

Why use Machine Learning?

Machine Learning Generalisation?

Project Aims and Objectives (500 words)

Aims

- Overview of how this project will answer the question/area proposed at end of introduction.

Project Plan (500 words)

Data Acquisition

Site Selection

Analysis

Feature Engineering

ML Model Setup and Training

ML Model Evaluation

ML Model Predictions

Tools

Interpretation

Gantt Chart

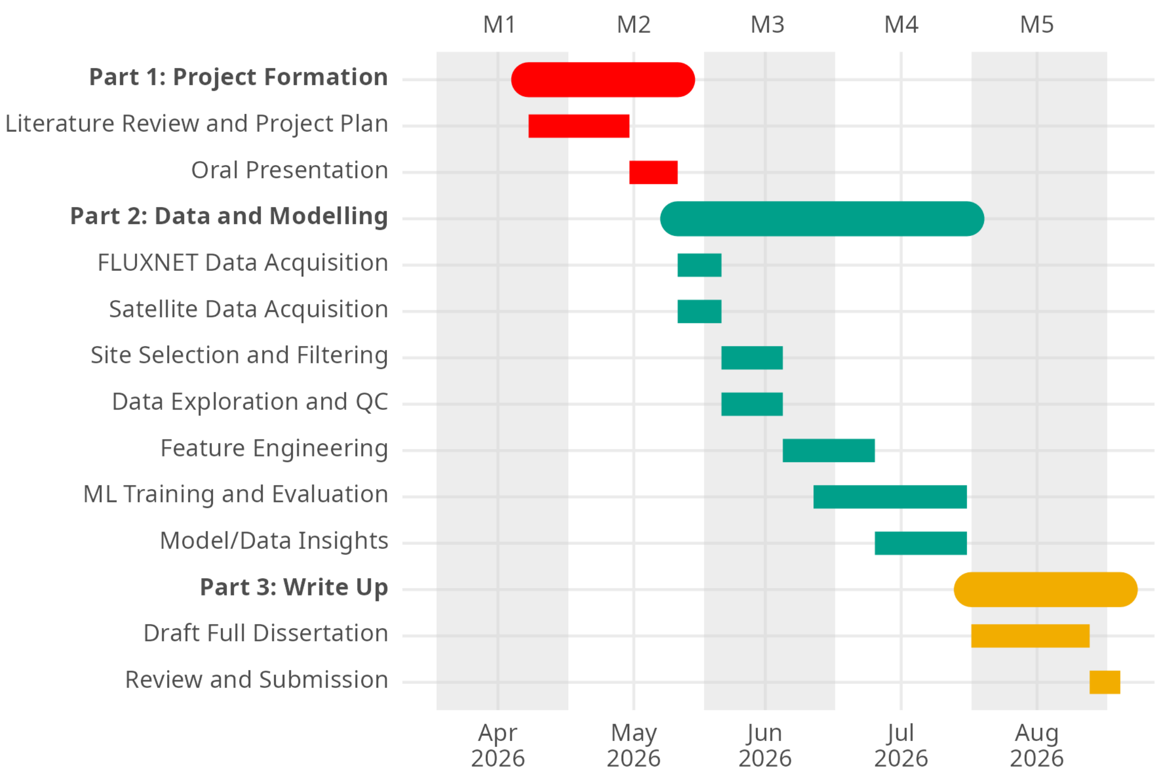

Figure 1: A proposed Gantt Chart for the duration of this project.

Risk Register (~ 250 words)

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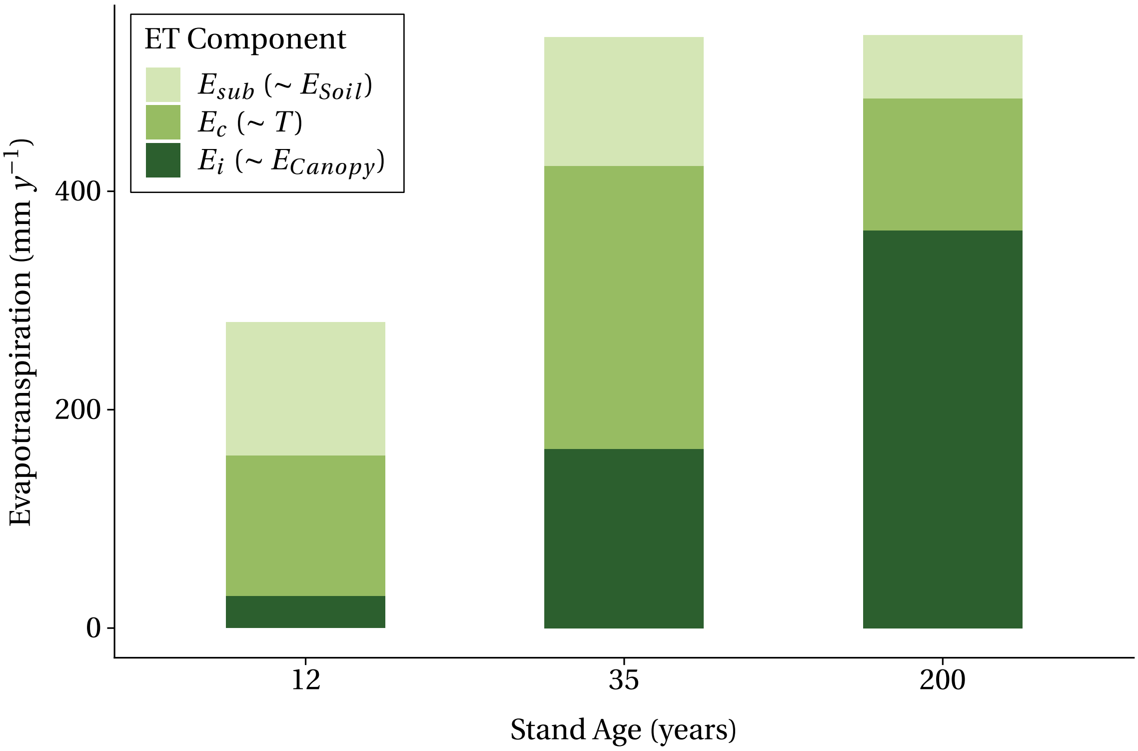

Figure 2: Components of annual evapotranspiration by stand age, redrawn from Oishi et al. (2020). Sub canopy evapotranspiration (E_{sub}), canopy transpiration (E_c), and canopy interception (E_i) correspond approximately to E_{Soil}, T, and E_{Canopy} in Equation 1, although E_{sub} additionally captures under canopy transpiration.

Method	Advantages	Disadvantages